Human Activity Recognition Through Ensemble Learning of Multiple Convolutional Neural Networks

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***Abstract*—Human Activity Recognition is a field concerned with the recognition of physical human activities based on the interpretation of sensor data, including one-dimensional time series data. Traditionally, hand-crafted features are relied upon to develop the machine learning models for activity recognition. However, that is a challenging task and requires a high degree of domain expertise and feature engineering. With the development in deep neural networks, it is much easier as models can automatically learn features from raw sensor data, yielding improved classification results. In this paper, we present a novel approach for human activity recognition using ensemble learning of multiple convolutional neural network (CNN) models. Three different CNN models are trained on the publicly available dataset and multiple ensembles of the models are created. The ensemble of the first two models gives an accuracy of 94% which is better than the methods available in the literature.**

***Index Terms*—Human activity recognition, ensemble learning, deep learning, convolution neural networks.**

I. Introduction

Human Activty Recognition (HAR) has developed its application in multiple fields and for a variety of tasks, including smart healthcare systems, automated surveillance and security. For the purpose of HAR, on-body sensors, such as tri-axial accelerometers, gyroscopes, proximity sensors, magnetometers, and temperature sensors, etc., are preferred for three main reasons; firstly, they omit the privacy issues posed by video sensors. Secondly, an entire body sensor network (BSN) allows for more accurate deployment of a signal acquisition system, and finally, they reduce the restrictions in place due to the environment and the stationary placements of cameras, as in video based datasets.

HAR has identified its crucial place in ubiquitous computing [1] because of the eased process of collecting data with embedded sensors, especially with the widespread use of smart phones in the last decade. Therefore, smart phones can essentially serve as a network of on-body sensors for data collection.

Previously, the process of feature extraction from these collected data points remained heuristic, hand-crafted, and task-dependent. The feature extraction process has been dependent on the usage of features that are purposefully designed and selected depending on the application. Statistical measures, such as the mean, and the variance, and transform coding measures, such as Fourier transforms, were extracted from the raw signals and subsequently used for classification methods. These methods posses the limitation of being bound to the specific classification tasks they are designed for. Manual selection of features also poses the problem of losing out on information from the raw signal [2].

With the advancements in deep learning, this process has been accelerated with unprecedented improvements, especially with the availability of more data and higher computation power. The feature selection process no longer needs to be task-dependent as deep learning models can extract features automatically and they can be applied to a range of classification tasks. With the provision of deep learning, it makes a logical flow to maximise the use of this state-of-the-art research for concentrated implementation of HAR. Therefore, the aim of this work is to develop a systematic process for a more streamlined feature representation for improved activity recognition through ensemble learning of CNNs [3].

The existing work in the field of HAR includes a very avid implementation of convolutional neural networks. Deep CNN in [4] adapts automated feature learning from raw inputs through its novel unification layer that concatenates feature maps at the end. A simpler CNN based approach is described in [5] where the model is trained for 3 different dataset subgroups extracted from the same dataset repository. A long short-term memory (LSTM) network based feature extraction approach along with a softmax classifier is utilized in [2] due to the effectiveness of LSTMs for repetitive tasks. The combination of recurrent neural network (RNN) and LSTM architecture, RNN-LSTM, has also been explored for this application since the dataset used is one-dimensional, similar to that of applications in natural language processing (NLP) for which RNN has produced the most promising results [6]. HAR has also been implemented through hybrid models of CNN and RNN-LSTM for improved performance in [7]. Such a hybrid model combines the best of both worlds as RNNs and LSTMs are able to recognize short range activities with greater accuracy due to better inference of time-order relationships

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between sensor readings. CNNs possess a better capability of inferring repetitive, long-term activities by more effectively learning features in recursive patterns.

An interesting approach towards HAR has been highlighted using deep belief networks (DBNs) through restricted Boltzmann Machines (RBMs) to ensure that the model doesn't overfit and the time consumption during the training process remains restricted [8]. Another hybrid approach for sequential human activity recognition is highlighted in [9]. It explores the use of an extreme learning machine (ELM), which is a feedforward network with selected hidden layers and nodes, which do not require tuning. The ELM is used as a classifier and is incorporated with a combined CNN-LSTM architecture as the feature extractor. The combination of convolutional operations with an LSTM for recurrent units is highlighted for effective sequential activity recognition.

Another LSTM-CNN architecture for HAR is encountered in [10]. In this architecture, the raw data collected by mobile sensors is fed into a two-layer LSTM followed by convolutional layers. In addition, a global average pooling layer (GAP) is also applied for reducing model parameters. Moreover, a batch normalization layer (BN) is also added after the GAP layer to speed up convergence. The model is tested with various HAR datasets with high evaluation accuracy. [11] introduces a method that takes into account the increased computational complexities and computing resource overheads tied with many deep learning approaches. This approach makes the task of HAR more efficient and manageable on mobile devices with lower computational resources. The method outlined in the paper uses feature extraction on multiple segments of the raw signal using temporal convolutions on the spectrogram domain of the data. This makes the learned features invariant to changes in various properties.

The efficacy of existing methodologies for HAR are contingent upon a wide variety of factors. With traditional deep learning approaches, the problems of overfitting and time consumption for model training are heavily present. These models also create a difficulty in comparison with HAR models that are developed, keeping in mind devices with lower computational complexities, such as mobile devices.

Here in this paper, we present an approach that is of forming three CNNs with one-dimensional (1D) convolutional layers for activity prediction on 1D HAR dataset. The 1D convolution layer creates a convolution kernel that is convolved with the layer input over a single spatial or temporal dimension. It then further goes to ensemble the three individual models together by taking their averages. This, ensemble learning as a multiple classifier system [12], improves the model performance and produces results better than those of individual models. We then further on create other possible ensembles of two models, i.e. of first and second, second and third, and, first and third models. Ensemble learning as a whole has shown an improvement in the performance of the model.

The rest of the paper is organized as follows: Section II presents the details of dataset, Section III describes the methodology, Section IV presents and discusses the obtained results, and Section V concludes the paper.

II. Dataset

The dataset used for the application of ensemble learning on Human Activity Recognition is the one provided by Wireless Sensor Data Mining (WISDM) Lab [13]. This dataset has been collected by 36 users each with a smart phone. The sensor used for collecting data is the embedded tri-axial accelerometer (x-axis, y-axis, z-axis). The samples are taken with a frequency of 20 Hz (20 samples/second). The dataset includes 6 daily activities as classes. The class distribution is as follows:

- Walking: 424,400 examples, 38.6%,
- Jogging: 342,177 examples, 31.2%,
- Upstairs: 122,869 examples, 11.2%,
- Downstairs: 100,427 examples, 9.1%,
- Sitting: 59,939 examples, 5.5%,
- Standing: 48,395 examples, 4.4%

This class distribution is plotted as shown in Figure 1.

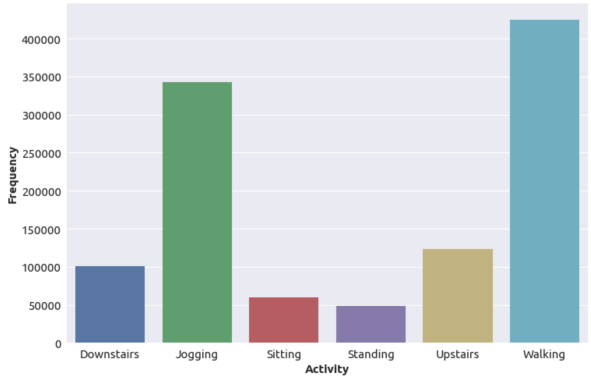


Fig. 1. Class distribution for WISDM dataset

First we pre-process the data such that it is normalised, that is to have zero-mean axis. The 3 axis graph for one activity (walking) is, as shown in Figure 2.

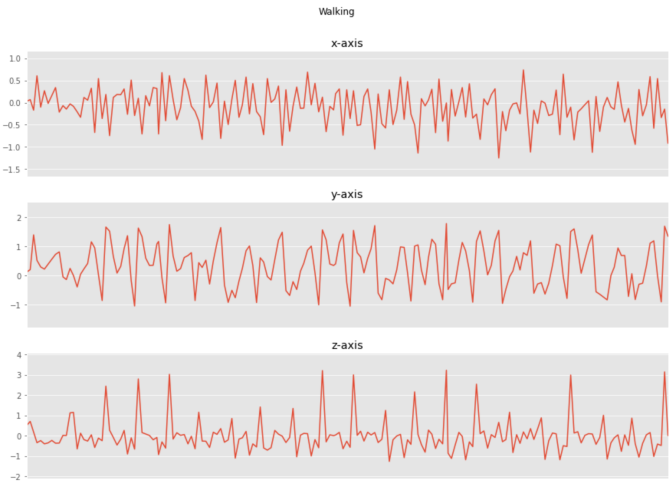


Fig. 2. Tri-axial data for a single class (walking)

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After pre-processing, we segment and shape the data such that it is compatible with 1D convolutional layers of the deployed CNNs. For segmenting the data, we first generate indexes of a fixed size window and move by half the window size. Then we generate fixed size segments and append the windowed data such that the dataset created has dimensions such as [total segments, input width and input channel] which is [24403, 90, 3] for our case.

III. Methodology

In this paper, we developed three different CNN based models, using which, a series of ensemble learning models were also created. Ensemble learning is a paradigm by which multiple models or learners are trained to solve the same problem. The advantage of this paradigm lies in its ability to generalize. It has the ability to boost the learning effect of learners that are weaker, leading to an improved performance output. The following sub-sections describe the overview and characteristics of the models.

A. First Model

This first model is inspired from the ConvPool-CNN-C architecture highlighted in [14]. It features a fairly common pattern of alternating convolution and max pooling layers. For this model, the 2D convolutions and max pooling operations were adapted into 1D operations with ReLU activation functions used for the convolutional layers. An additional feature introduced is the use of dropout regularization after the application of each convolution filter to reduce the tendency of overfitting the training data. This method yields an improved response as opposed to bare utilization of convolution and max pooling layers. Fully connected layers are also used in this model as opposed to global average pooling as used in ConvPool-CNN-C. The first fully connected layer towards the end of the model utilizes a *tanh* activation for normalization followed by a *softmax* function in the final dense layer to optimize for the classification scenario at hand. Figure 3 shows the architecture of the model.

B. Second Model

The second model used is adapted from the ALL-CNN-C, mentioned in [14]. The main difference between this model and the first one is the use of convolutional layers in place of max pooling layers and hence, the name, ALL-CNN (all-convolutional neural network). For this model, a number of 1D convolutional layers with ReLU activation were used with stride 2, in place of the max pooling layers. Figure 4 shows the architecture of Model 2.

C. Third Model

The third model is inspired from the 'network in a network' model, from [15], that uses a series of convolution layers with 1x1 kernels to reduce the number of parameters for optimization and reduce training time. Dropout regularization was also introduced to improve upon accuracy. The model architecture is shown in Figure 5.

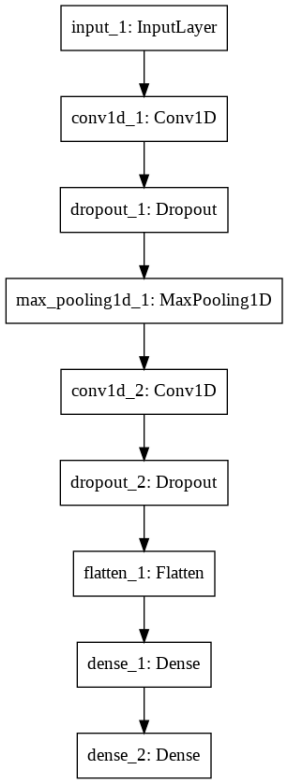


Fig. 3. Model 1, inspired from the ConvPool-CNN-C architecture. It utilizes alternating 1D convolution and max pooling layers with a fully connected layer.

D. Ensemble of 3 models

In the next step, we created ensemble of the three models. It uses the same input layer that is shared between each of the three models and, at the end, computes the average of the outputs of the three models to produce a mean output. While there are different methods of building an ensemble model, here we used a stacking ensemble pattern whereby the average of each of outputs of the CNN architectures used in the model is computed and output. The only training done is for the individual models and no training is required for the ensemble model itself. Figure 6 shows an overview of the three CNN ensemble model. Apart from an ensemble of the three models, pairwise ensemble models were also produced, with each of the three CNN models paired with each other.

IV. Results and Discussion

We trained and tested our models and their ensembles using (WISDM) dataset [13]. Figures 7 and 8 show plots of the training accuracies and losses of the three CNN models. After about 5 epochs, the accuracies crossed the 90% threshold and approached more than 99% accuracy over the course of the 50 epochs. The validation accuracies, however, lay between 70 - 80% with a higher degree of variation observed as compared to the training procedure.

The ensemble model, on the whole, exhibited much higher training accuracies and was able to translate that to a better

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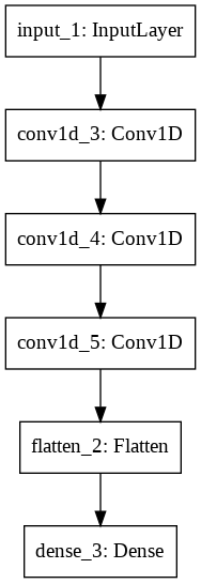


Fig. 4. Model 2, adopted from the ALL-CNN-C architecture. This model utilizes all convolution layers and does not use any max pooling layers.

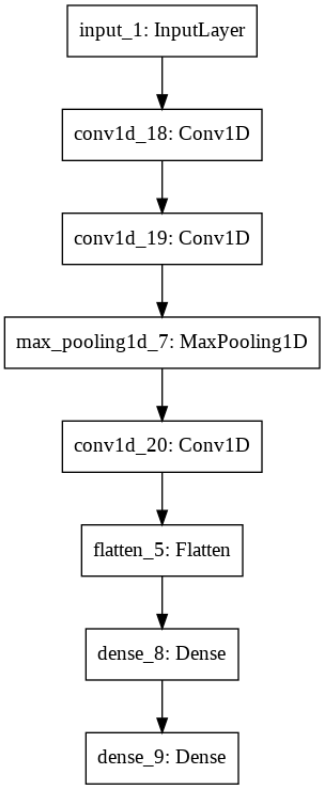


Fig. 5. Model 3, partially inspired by the Network in a Network (NIN) model. It utilizes a series of 1D convolution layers, a max pooling layer and fully connected layers.

evaluation accuracy. Table 1 shows a summary of the evaluation accuracies of each of the CNN models vs. the ensemble models. As evident from Table 1, the ensemble models tend to produce higher degrees of evaluation accuracy as opposed to

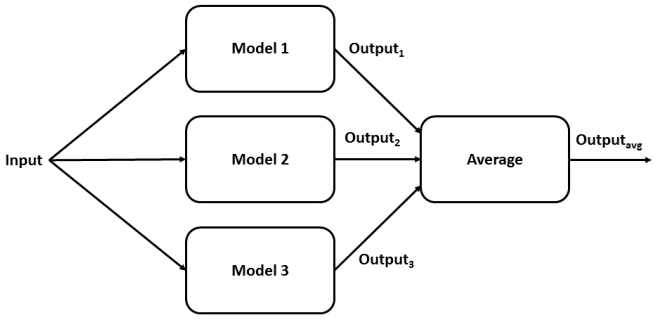


Fig. 6. Ensemble Model

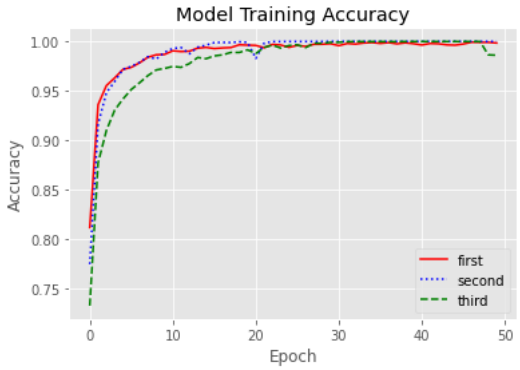


Fig. 7. Training Accuracies for Individual Models

individual CNN models. Notice particularly the results where the third model is included in the ensemble and its accuracy is increased (approx. 93%) compared to the 90% accuracy of the third model alone.

Throughout the literature, different non-ensemble architectures were encountered which yielded particular accuracies and losses. The LSTM-RNN based HAR model in [6] yielded a 90% accuracy after 200 epochs. The simple CNN based approach in [5] yields a maximum recognition rate of 92.5%. The multiclass SVM approach mentioned in [8] yields an 82% accuracy rate. The inverted pyramid architecture based convnet plus multilayer perceptron in [16] was able to achieve an accuracy of 93.75%. Compared to these approaches, our pairwise ensemble model, Paired Ensemble (1,2), was able to achieve an accuracy of above 94%. In general, the ensemble models developed were able to yield lower evaluation losses

TABLE I
MODEL ACCURACIES

Model	Accuracy (%)
First	93.08
Second	93.55
Third	90.93
3-Model Ensemble	93.66
Paired Ensemble (1,2)	94.01
Paired Ensemble (2,3)	93.04
Paired Ensemble (1,3)	93.08

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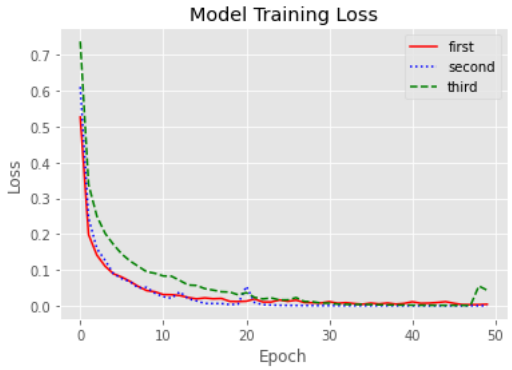


Fig. 8. Training Losses for Individual Models

as compared to the individual deep learning models developed.

V. Conclusion

We presented three CNN based models as well as their ensembles for WISDM dataset of HAR. It was found that the performance of the ensemble model is better than that of individual models. One of the ensemble model performed better than the methods in the literature. In the dataset we used, we see a class imbalance such that we have 38% samples for walking class but hardly 5% for sitting and standing. In future, the results might be improved even more, if we can remove the class imbalance from dataset. Moreover, currently an ensemble of average of the three models is created but for further exploration, a future direction can be performing weighted ensemble learning such that the best performing model has the most effect in the ensemble. Furthermore, we can explore the area of ensemble learning for a hybrid model, that is, ensemble learning of CNN and RNN models.

References

[1] T. Plötz, N. Y. Hammerla, and P. L. Olivier, "Feature learning for activity recognition in ubiquitous computing," in *22nd International Joint Conference on Artificial Intelligence*, 2011.

[2] Y. Chen, K. Zhong, J. Zhang, Q. Sun, and X. Zhao, "LSTM networks for mobile human activity recognition," in *International Conference on Artificial Intelligence: Technologies and Applications (ICAITA 2016)*, 2016.

[3] L. K. Hansen and P. Salamon, "Neural network ensembles," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 12, no. 10, pp. 993–1001, 1990.

[4] J. Yang, M. N. Nguyen, P. P. San, X. L. Li, and S. Krishnaswamy, "Deep convolutional neural networks on multichannel time series for human activity recognition," in *24th International Joint Conference on Artificial Intelligence*, 2015.

[5] M. Panwar, S. R. Dyuthi, K. C. Prakash, D. Biswas, A. Acharyya, K. Maharatna, A. Gautam, and G. R. Naik, "CNN based approach for activity recognition using a wrist-worn accelerometer," in *2017 39th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, 2017, pp. 2438–2441.

[6] S. W. Pienaar and R. Malekian, "Human activity recognition using LSTM-RNN deep neural network architecture," in *2019 IEEE 2nd Wireless Africa Conference (WAC)*. IEEE, 2019, pp. 1–5.

[7] J. Wang, Y. Chen, S. Hao, X. Peng, and L. Hu, "Deep learning for sensor-based activity recognition: A survey," *Pattern Recognition Letters*, vol. 119, pp. 3–11, 2019.

[8] M. M. Hassan, M. Z. Uddin, A. Mohamed, and A. Almogren, "A robust human activity recognition system using smartphone sensors and deep learning," *Future Generation Computer Systems*, vol. 81, pp. 307–313, 2018.

[9] J. Sun, Y. Fu, S. Li, J. He, C. Xu, and L. Tan, "Sequential human activity recognition based on deep convolutional network and extreme learning machine using wearable sensors," *Journal of Sensors*, vol. 2018, pp. 1–10, 09 2018.

[10] K. Xia, J. Huang, and H. Wang, "LSTM-CNN Architecture for Human Activity Recognition," *IEEE Access*, vol. 8, pp. 56 855–56 866, 2020.

[11] D. Ravi, C. Wong, B. Lo, and G. Yang, "Deep learning for human activity recognition: A resource efficient implementation on low-power devices," in *2016 IEEE 13th International Conference on Wearable and Implantable Body Sensor Networks (BSN)*, 2016, pp. 71–76.

[12] M. Woźniak, M. Graña, and E. Corchado, "A survey of multiple classifier systems as hybrid systems," *Information Fusion*, vol. 16, pp. 3–17, 2014.

[13] J. R. Kwapisz, G. M. Weiss, and S. A. Moore, "Activity recognition using cell phone accelerometers," *ACM SigKDD Explorations Newsletter*, vol. 12, no. 2, pp. 74–82, 2011.

[14] J. T. Springenberg, A. Dosovitskiy, T. Brox, and M. Riedmiller, "Striving for simplicity: The all convolutional net," *arXiv preprint arXiv:1412.6806*, 2014.

[15] M. Lin, Q. Chen, and S. Yan, "Network in Network," *arXiv preprint arXiv:1312.4400*, 2013.

[16] C. A. Ronao and S.-B. Cho, "Human activity recognition with smartphone sensors using deep learning neural networks," *Expert Systems with Applications*, vol. 59, pp. 235 – 244, 2016. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0957417416302056

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